Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/06/2005
PI Name:	van der Woerd, Mark Ph.D.		
Project Title:	The Master Switch for Bone Formation: Structural Studies of RunX2		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	80523	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	03/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	1/16/07--Received extension to 3/31/2007, per A. Chu 9/29/06--Received NCE to 12/31/2006, per A. Chu. 07/2006--PI moved from BAE Systems to Colorado State U and transfered project to Colorado State U; project with BAE Systems to be closed out (per A. Chu)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lian, Jane (University of Massachusetts Medical School) Snell, Edward (Hauptman-Woodward Medical Research Institute) Ferree, Darren (BAE SYSTEMS Analytical Solutions) Karr, Laurel (NASA/MSFC) Stein, Gary (University of Massachusetts Medical School) Montecino, Martin (University of Massachusetts Medical School)		
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Task Description:	Runx2, which is also called the Core Binding Factor a1 (Cbfa1), is the pivotal molecule for bone formation – the master switch. It regulates osteoblast growth and differentiation at the level of gene transcription, performing these regulatory functions by activation and repression of genes important for bone formation in the osteoblasts. While Runx2 binds to the gene (DNA) with a domain near the N-terminus, the C-terminal region is essential for biological activity. The functional activity of the C-terminus is influenced by a nuclear matrix targeting signal that localizes Runx2 to subnuclear foci for the assembly of transcription factor complexes with a host of interacting co-regulating proteins. We postulate that a significant conformational change occurs within the C-terminal domain of Runx2 upon specific binding of a co-regulatory protein. We hypothesize that the structural changes will differ, dependent on which factor is bound. We are determining both the structure and function of parts of the native and mutant RunX2 molecules, so we will better understand how molecular structure leads to biological function. One of our specific aims is to resolve by X-ray crystallography the three-dimensional structures of the C-terminal domain of wild type and mutant Runx2 molecules, in the absence and presence of regulatory factors. A second specific aim is to experimentally determine the functional activities of Runx2 wild type and point mutants and to identify the mechanisms by which the interactions between Runx2 and other molecules regulate osteoblast differentiation. Significance for NASA: Bone loss occurs at a rate up to 1.5% per month in astronauts while they reside in a reduced gravity environment. It occurs due to an imbalance between bone formation (osteoblast activity on the cellular level) and bone resorption (osteoclast activity). Research on bone loss in long-term reduced gravity missions conducted on astronauts has shown that bone formation is reduced and resorption is unaltered or increased. Our research will help to explain how the activity of the osteoblasts is regulated at the molecular level. Greater understanding of osteoblast activity will lead to future opportunities to better regulate bone formation and therefore reduce bone loss in space and on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a recognized disease that affects millions of people, particularly the elderly. It is responsible for hundreds of thousands of fractures annually in the United States alone. Osteoporosis is caused by an imbalance between bone formation and resorption. Each process may be used to restore the balance, for example by anti-resorptive agents or by stimulation of bone formation. The results from our research will contribute to a better understanding of the processes that occur during bone formation. Increasing bone formation is one of the avenues to restore a healthy balance and avoid osteoporosis.
Task Progress:	Runx2 interacts with co-regulatory proteins that change its functional activity. One such protein is called YAP, a mediator of signaling pathway important for bone formation and bone resorption.Our hypothesis is that the transcription factor RunX2 will show a conformational change upon binding the protein YAP. To prove this, we will need to determine and study the structure of the (relevant part of) RunX2 protein, in the presence and absence of either the entire protein YAP, or at least YAP's binding domain. For this purpose we need a large quantity of the purified RunX2 and YAP proteins each, typically in the 10-100 mg range. We have over-expressed a protein in E. coli from DNA with a sequence confirmed to be identical to a relevant part of RunX2 and with the correct molecular weight. Purification of this protein and further identification by means of Western blots are in progress. The production of YAP protein domains in a bacterial host is in preparation. The functional assays of RunX2 in living cells have shown the influence of RunX2 itself - it promotes osteoblast activity - and the influence of RunX2/co-regulatory binding proteins such as YAP, which represses the activation of target genes related to bone formation. This year we have studied the RunX1 molecule, very closely related to RunX2 and known to be important in hematopoietic cells lines. We found that it has a pronounced influence in the functional activity of the osteoclast, thus directly affecting bone resorption. This discovery shows that the RunX family of proteins is important to both bone formation and bone resorption. The studies of how co-regulatory factors bind to the RunX family of proteins are even more relevant than was thought before.
Bibliography Type:	Description: (Last Updated: 08/23/2007)
Articles in Peer-reviewed Journals	Saltman LH, Javed A, Ribadeneyra J, Hussain S, Young DW, Osdoby P, Amcheslavsky A, van Wijnen AJ, Stein JL, Stein GS, Lian JB, Bar-Shavit Z. "Organization of transcriptional regulatory machinery in osteoclast nuclei: Compartmentalization of Runx1." J Cell Physiol. 2005 Sep;204(3):871-80. PMID: 15828028 , Sep-2005
Articles in Peer-reviewed Journals	Pratap J, Javed A, Languino LR, van Wijnen AJ, Stein JL, Stein GS, Lian JB. "The Runx2 osteogenic transcription factor regulates matrix metalloproteinase 9 in bone metastatic cancer cells and controls cell invasion." Mol Cell Biol. 2005 Oct;25(19):8581-91. PMID: 16166639 , Oct-2005